OUTBACK ROBOTS

BUILD. LEARN. LEAP AHEAD.

Lesson 1 - Meet Blossom:

An Introduction to Social Robots

HAVIC LAB
HUMAN ARTIFICIAL + VIRTUAL INTERACTIVE COGNITION

MONTEBELLO FOUNDATION

Flinders University

Teacher-only slide – Instruction Guide

4 Types of Slides

Student Content

- Slides in this colour are the ones that will be shown to the students.
- They contain little text and often have images illustrating the concepts.

Teacher-only slide – Instruction Guide

Instruction Guides

- The Instruction Guide slides give an overview of how to instruct the next slide activity.
- They describe what to say or ask about the following 'student content' slide, what the class is meant to be doing, and what answer the class is expected to get to.
- The regular text on these slides is teacher guidance, i.e. notes from us to you. Some of these slides include "*teacher prompts which are written in italics and inside of quotation marks*". These are example instruction scripts with optional wording for the next slide that you can read aloud, adapt to your style or use a different instruction altogether.

Teacher-only slide – Lesson Overview

Lesson Overview Slides

- Lesson Overview slides appear at the beginning of each lesson.
- They are meant for the teacher to get a sense of what the lesson will be about and what needs to be prepared beforehand.
- Each lesson's slide deck contains:
  - a lesson preparation slide (what needs to be prepared before the lesson)
  - slides with the learning objectives and student success criteria for the lesson
  - a lesson overview slide with the different topics/activities covered in that lesson and how much time will approximately be spent on each of them

Teacher-only slide – Content Extension

Content Extension Slides

- Slides in this colour are meant as teacher-only content extension slides.
- They give teachers a deeper understanding of some of the concepts so they feel confident answering questions
- Sometimes, they offer examples, analogies, and common misconceptions to watch out for.

Teacher-only slide – Lesson Overview

Lesson Preparation

1. Before you start:

- Build your own Blossom to get familiar with the process and use it for demonstration purposes (‘Teacher’s Blossom’)
  - Contact the study team in advance in case you run into any issues
- Familiarize yourself with the slides and activities in this lesson.
- You can print off the teacher slides or add any notes/parts of the example scripts to the notes on the student slides. This way, you can see them when in presenter mode.
  - Then hide the teacher slides, so they don’t get shown when you present.
- We recommend to go through the slides in presenting mode, so that you also become familiar with the animations and order in which the different items appear.

Teacher-only slide – Lesson Overview

Lesson Preparation

2. Classroom Setup

- Plan student groups for the hands-on activities throughout the program
  - Our recommendation: 3 students per Blossom kit, select groups such that
    - ability is roughly equally distributed between the groups, so groups progress at a similar pace, and
    - conflict and distraction are minimized within the groups.
- Materials needed: Blossom kits, pre-assembled Teacher's Blossom, one student device (laptop or tablet) per group

3. Right before the lesson:

- connect to the Teacher's Blossom's WiFi, wait 5 minutes, then open the interface at https://192.168.90.1:8080

Teacher-only slide – Lesson Overview

Lesson 1 - Overview

1. **Introduce Blossom** - 5 min
2. **Outback Robots Program & Study Overview** - 5 min
3. **Robots** - 20 min
    3.1 Where can robots be found?
    3.2 Social Robots
    3.3 What is (not) a robot?
4. **Parts of a robot** - 45 min
    4.1 What parts does every robot need?
    4.2 Investigating the pieces of Blossom and their functions
    4.3 Making a motor move
5. Optional extension: **When should robots (not) be used?** - 15 min
6. **Questionnaire** - 5 min
7. **Lesson summary -** 5 min

Total: XX min (+ 15 min extension)

Teacher-only slide – Lesson Overview

Lesson 1 – Learning objectives

After this lesson, students will be able to:

- explain where robots appear in everyday life,
- understand that there are different types and categories of robots,
- distinguish robots from other machines,
- identify the essential components of a robot (sensors, controller, motors, wires) and their specific functions,
- discuss helpful and harmful uses of robots across different settings.

Lesson 1 - Student success criteria

- I can explain what the key features of robots are and what makes them different from other machines.
- I can name 3 types of robots and what they are being used for.
- I can identify the main components that robots require to function.
- I can describe what the functions of the motors, computer and wires inside a robot are, and recognize them among the pieces of Blossom.
- I can discuss when robots should or shouldn’t be used by giving 2-3 example scenarios and give reasons on why using robots would be helpful or harmful in this context.

Teacher-only slide – Instruction Guide

Introduction Suggestion

- Announce that you brought a special guest for today's lesson.
- Showcase the assembled teacher's Blossom robot, make it move using the interface.

Teacher-only slide – Instruction Guide

Program Overview

- **What is Outback Robots?**
  Use the Program Overview slide to explain that over the next 8 lessons, your class gets the chance to take part in a project by Flinders University through which students will learn about robots and AI. They will work with a robot called Blossom and do hands-on activities in teams. This also includes building their own robot.

- **Why are we doing it?**
  Robots and AI are increasingly becoming a part of everyday life. This program helps you understand how they work and how to make good choices while using them.

- **What the "study" means?**
  The scientists who created the lessons want to know what they are like for students and teachers so they can improve them. To help them, you will all fill out a few questions at the end of every lesson.

Teacher-only slide – Instruction Guide

Program Overview

Artificial Intelligence

Understanding AI | Using AI | Evaluating AI

1. **Meet Blossom:** An Introduction to Social Robots
2. **Sensing the World:** From Signals to Data
3. **Coding (with) Blossom**
4. **What is AI?** Algorithms and Predictions
5. **Ethics of AI:** Bias, Fairness and Impact
6. **Design for Good:** AI, Society and Sustainability
7. **Chat Smart, Stay Safe:** Generative AI, LLMs and Digital Safety
8. **Show, Tell, and Reflect:** Presenting Your Robot

Building Blossom | Interacting with Blossom

Social Robotics

Program Overview

OUTBACK ROBOTS

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Artificial Intelligence

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Building Blossom | Interacting with Blossom

Social Robotics

Program Overview

1. Meet Blossom
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4. What is Artificial Intelligence?
5. Chat Smart, Stay Safe
6. Ethics of AI
7. Design for Good
8. Show, Tell, and Reflect

Building Blossom

Interacting with Blossom

Teacher-only slide – Lesson Overview

The Research Team

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Sophia Jaeger

Curriculum and Material Development

Ben will be available remotely to provide support during the lessons for both you and students.
You can use the next slide to introduce him to the class.

Online support – Ben Kelly

Ben Kelly

Quick intro to what Ben will do in the lessons and that he is available to help with anything the students struggle with

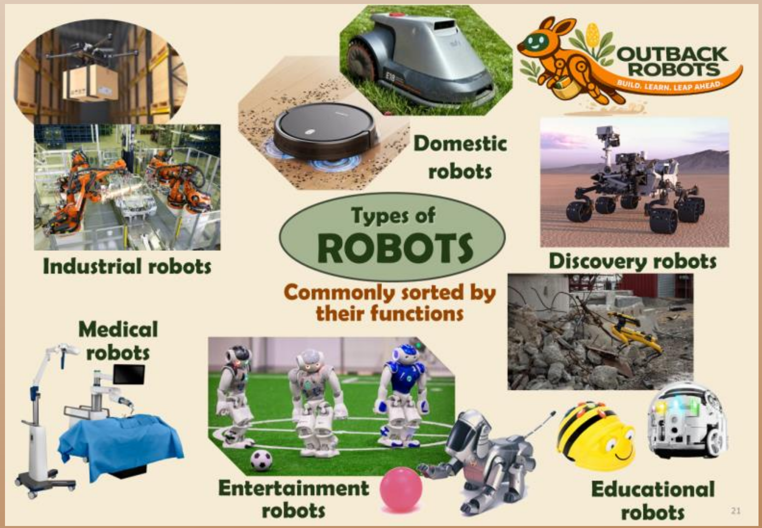
Teacher-only slide – Instruction Guide

Robots – Hook & Brainstorm

- **Example Script**:
  - *"Where have you already seen robots being used? What did they look like? What other types of robots do you think exist?"*
- **Instruction Note**:
  - Students might already have strong robot images (often humanoid robots from movies)
    - if this is the case, you can use this result as the basis for further discussion later in this lesson
    - there is also a teacher extension slide about humanoid robots later in this slide deck (see screenshot on the right)

Teacher-only slide – Content Extension

Robot Types

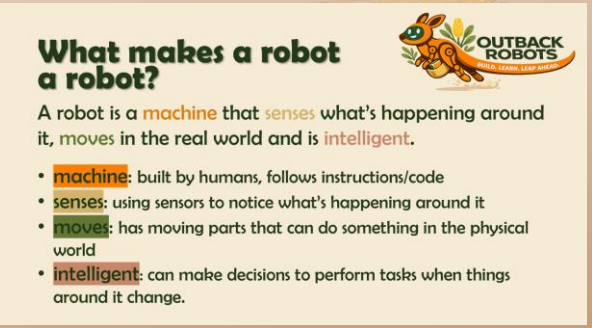


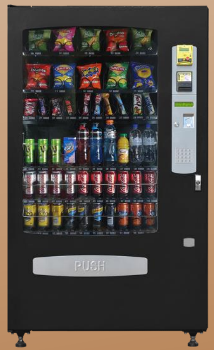
- A key learning point in this lesson is that the definition and distinguishing criteria of 'robots' is fluid (because technology is always changing) and sometimes ambiguous
- Robots can be **categorized based on** their **applications/functions** (see "Types of Robots" slide below), their **degree of movement** (stationary vs mobile), or their **autonomy** (f.i. vacuum robot has lower autonomy, while a robot assistant has more).
- The types of robots introduced in this lesson are categorized by function. This is a common way to sort robots into groups.

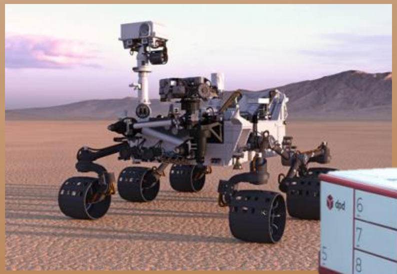
Teacher-only slide – Content Extension

Autonomy

- You can extend this lesson or discussion by introducing students to the concept of robot ‘autonomy’.
- Autonomy = How much a machine can do by itself without a human controlling every step (i.e., how much help a machine needs from a human)
- Can extend on this point following the discussion of robot “Intelligence” on the “What makes a robot” slide







no autonomy

Toy car
moves only when it is pushed

low autonomy

Microwave, Vending Machine
performs the same action every time a button is pushed

medium autonomy

Robot vacuum
can handle slightly changing environments, avoids furniture and returns to charger

high autonomy

Space rover, Delivery robot
is able to deal with changing environments, navigates, avoids obstacles, plans routes

Autonomy comes on a spectrum/range

Teacher-only slide – Instruction Guide

Robot Types

- **Example Script:**
  - *“Robots come in many different shapes and are used across a range of applications.”*
- **Instruction Note:**
  – Present the different robots on the next slide and what they are for:
    - ***industrial:*** *perform repetitive manufacturing tasks (e.g., assembly in factory, warehouse drone)*
    - ***medical****: e.g., robotic tools (e.g., surgical arm); social robots for lifting patients from beds*
    - ***entertainment:*** *for fun and playing games (e.g., robot football tournaments, robot pet*
    - ***education:*** *in schools to teach students (e.g. about coding)*
    - ***discovery****: search and rescue, space missions*
    - ***domestic:*** *help with chores around the house (e.g. vacuum cleaner, lawn mower)*

  – On the next slide, the images of robots appear by category (with animations).

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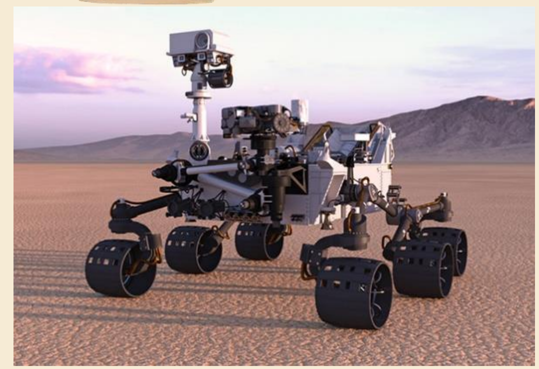
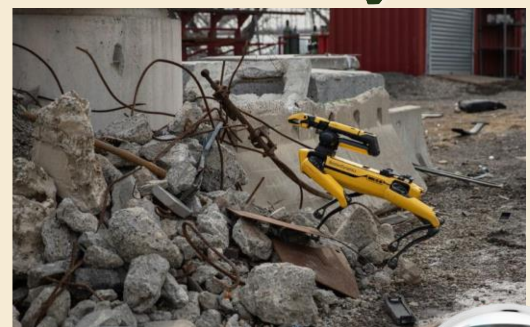
Industrial robots

Domestic robots



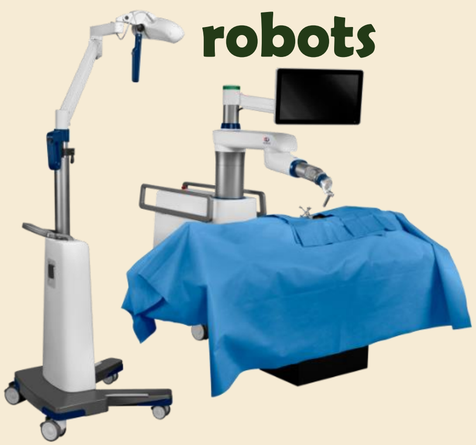
Discovery robots

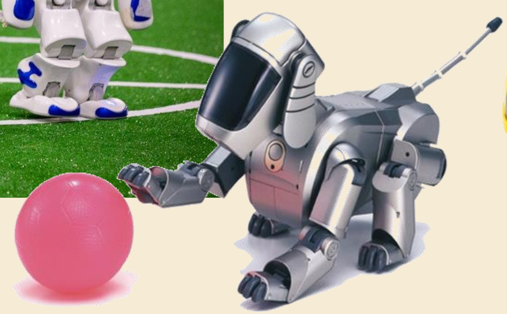


Types of ROBOTS

Commonly sorted by their functions

Medical robots





Entertainment robots

Educational robots

Teacher-only slide – Instruction Guide

Social Robots

- **Example Script:**
  - *"Another type of robots are social robots.*
    - *These are robots that can have social and emotional communication abilities.*
    - *They can often talk, gesture or express emotions. This can be done through movement (like in Blossom), facial features (many robots on the next slide) or speech.*
  - *Social robots are used for various applications and in different settings such as in education, healthcare, as assistant robots, or in customer service."*

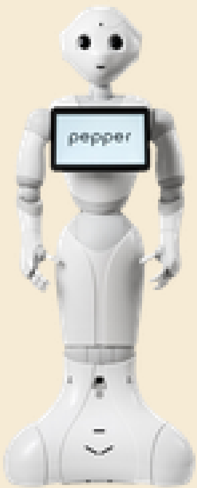
Social Robots

Robots that can have **social** and **emotional communication** abilities (e.g., talk, gesture, show emotion)

Education **Healthcare** **Assistant** **Customer service**





These robots can sometimes look human-like (e.g., have faces) but this is not a requirement for social robots.

Teacher-only slide – Content Extension

Humanoid, zoomorphic and mechanoid robots

- There are social robots that are humanoid, meaning they look like (and often behave like) humans. Some social robots are zoomorphic. They look like animals. But social robots can also be mechanoid.
- The Cozmo robots is a good example for this. It looks neither like a human nor an animal but still has social and emotional communication abilities.

Cozmo robot

Teacher-only slide – Instruction Guide

Humanoid Robots

- Introduce the concept of humanoid robots
  - Social Robots can be humanoid (robots that look like humans, often mimicking human movement or behaviour)
- Use the previous slide about social robots and have students identify which of the depicted robots are humanoid and what features they share with humans.
  - *"The extent to which a robot is humanoid varies. Some of the robots slightly resemble human features (e.g. the Marty robot), while others look and behave a lot more human-like."*

Marty robot

vs

Humanoid robot

Teacher-only slide – Instruction Guide

What do all robots have in common?

Instruction Note:

- Still using the previous slide with the different types of robots, let **students brainstorm** what all the robots have in common and what makes them different from other machines (like a microwave or a smartphone).
- Then go to the next slide: **What makes a robot a robot?**
  - Introduce the rule (**robot is a machine that senses, moves in the real world, and is intelligent**)
  - **Then say**:

    "It can be tricky to decide when a machine can be seen as intelligent. It does not equal "thinking like a human". Rather, it means the robot can use sensor information and human-written code (instructions) to choose a suitable action.

    This decision-making can be autonomous (can be done on its own without a human controlling every step), or remote-controlled (a human decides each action, the robot still acts but has little or no autonomy)."

What makes a robot a robot?

A robot is a machine that senses what's happening around it, moves in the real world and is intelligent.

- machine: built by humans, follows instructions/code
- senses: using sensors to notice what's happening around it
- moves: has moving parts that can do something in the physical world
- intelligent: can make decisions to perform tasks when things around it change.

Teacher-only slide – Instruction Guide

Is it a Robot?

Instruction Notes:

- Discuss with the students for the different images if each of them is considered a robot. Why (not)? (possible answers & considerations on the next slide)

Then explain:

- *"It's not always easy to decide, and there is ambiguity on what a robot is. There are some machines that are borderline, meaning some people consider them a robot while some don't. It's often a question of whether they can be seen as having enough intelligence or autonomy.*
- *The definition of what robots are is likely to keep changing as robots and AI technologies become more advanced."*

Teacher-only slide – Instruction Guide

Is it a robot?

Vending machine can be considered a (very simple type of) robot

✓ machine

✓ moving parts

✓ notices/sensor

yes buttons (very simple type of sensor)

? slightly intelligent, low autonomy

moves to correct item on press of a button

Vacuum cleaner (robot)

✓ machine

✓ moving parts

✓ notices/sensor

detect objects & walls, some even detect dirt

✓ intelligent, medium autonomy

cleans by itself and changes direction with obstacles, but still considered only partly autonomous since limited decision-making

Virtual assistant (not a robot)

✓ machine

X moving parts

cannot move and perform any actions in the physical/real world

✓ notices/sensor (microphone)

✓ intelligent, high autonomy

interacts socially (speaks, answers questions)

Printer can be considered a (very simple type of) robot

✓ machine

✓ moving parts

✓ notices/sensor

detects when low on ink, when paper jams, when human presses buttons

? slightly intelligent, low autonomy

performs an action only when initiated by a human, but stops when it detects an error

Is it a robot?

A robot is a machine that notices, thinks, and moves.

vending machine

vacuum cleaner

printer

virtual assistant (Alexa, Echo & Co)

Teacher-only slide – Instruction Guide

Working with Blossom

Instruction Note:

- Before handing out the kits, discuss with the students what them working with their Blossom and its parts should look like.
- You can do this using your school's values (e.g. how to handle pieces responsibly, be respectful towards others, be safe when building, ...). We have listed some options on what to talk about on the next slide.

- Further, we recommend playing the Blossom instruction video BEFORE handing out Blossom kits.
- Remind students which parts of the kit are to be used in this lesson.
- Instruct students not to unpack entire kit, which is organised to help them follow in subsequent lessons.
- We recommend to designate a student in each group responsible for kit.
- Then **explain the robot kits** to students.

Teacher-only slide – Instruction Guide

Working with Blossom

Our recommendations on what to mention:

- Be gentle with the pieces.
  - *“Some of the pieces can break if not handled gently. If you want your robot to work in the end, you need to handle the pieces with care so that nothing breaks unnecessarily.”*
- Do not plug or unplug any wires.
  - *“If the wrong wires touch or the Raspberry Pi is unplugged at the wrong time, this can impact its performance.”*
- Pieces stay in the kit (unless explicitly told otherwise).
  - *“Some pieces are quite small and can easily get lost when taken out of the kit but not used.”*
- No food or drinks near it.

- **General rule**: Do not do anything to Blossom that you wouldn’t do to a smart phone or laptop.

We have prepared a slide on basic rules (next slide). Feel free to hide or edit to include other class rules.

Working with Blossom

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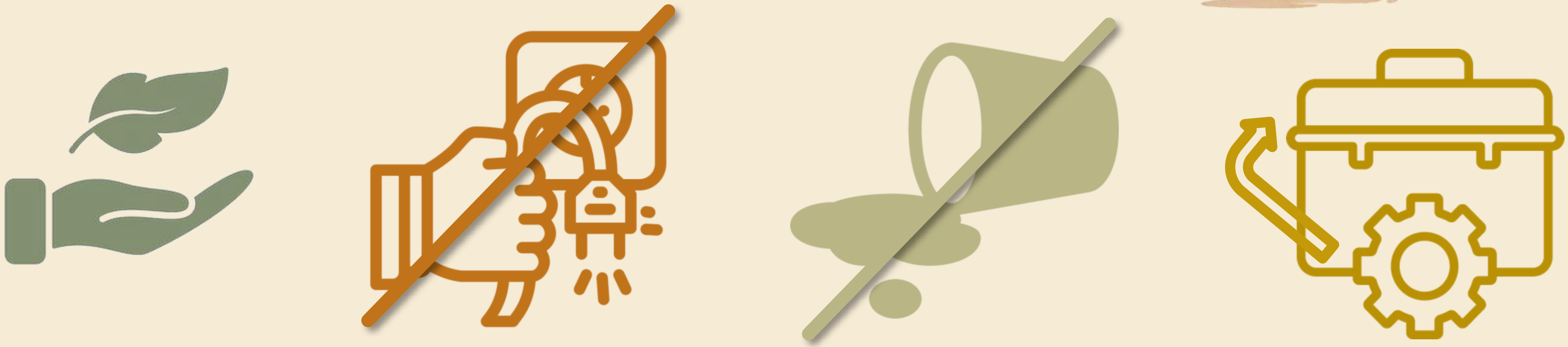


- Be gentle with the pieces.
- Do not plug or unplug any wires.
- Leave the Blossom case closed.
- The pieces stay in the kit (unless instructed otherwise)
- No food or drinks next to Blossom.

Hands-on Activity Instruction

Instruction note:

- Arrange students into the groups you decided on before the lesson. Let them change seats to sit with their group.
- Let one students from each group come to you and get a Blossom kit.
- Once all groups have their kit, let them open it and take out only the paper checklist. Make them write their names on it and place it in front of them.
- In the next section, students will need to find certain items in the kit (motors, computer case, sensor). Highlight again that they are not to take out any other parts and should return each piece to where it came from (the kit is labelled).

Teacher-only slide – Instruction Guide

Blossom Building – the components

Example Script:

- *"A robot is made up of many different components. What parts do you think every robot needs and would not be able to function without?"*

Instruction Note:

- Some common and important robot components that can come up in the conversation:
  – **power supply**: source of energy (can be electrical, but also other options like pneumatic or hydraulic)
  – **control system**: computer that processes input, executes code and controls other parts such as motors (like human brain)
  – **software and programming**: code running on the control system, instructions on what to do and when to do it (like human thoughts & skills)
  – **sensors**: take in information from the robot's environment (like human sensory organs)
  – **motors**: produce movement (like human muscles)
  – **mechanical structure**: physical frame/body for movement and durability (like human bones)
  – **wires**: connect different parts, transmit power or signals (like human nerves and brain cells)
- It is not a problem if students are not able to name all of the above. You will go through the most important ones in a bit more detail in the following slides.

Teacher-only slide – Instruction Guide

Servo Motor

Instruction Note:

- Let each student find a tower motor from their kit. (There are 3 per kit.)
- Let them inspect it, think about what the piece is for, and how it performs this task (~ 2 min). Then ask them for their thoughts.

Answer:

– *"This is the motor. The motor creates movement.*

– *It gets powered through the wires. When it receives a signal to move, the motor turns, which turns the wheel it's attached to. A string is attached to the wheel on one end and to Blossom's head on the other. When the motor & wheel turn, the string gets wrapped around the wheel, making it shorter. This pulls on the head and makes it move."*

Instruction Note:

– Demonstrate this on the assembled Teacher Blossom.

Find this piece (3x)

- SG90 tower -

What could it be for?

Motor

- makes the robot move

How does it do that in the Blossom robot?

out of the box

in the Blossom kit

in assembled robot

Teacher-only slide – Instruction Guide

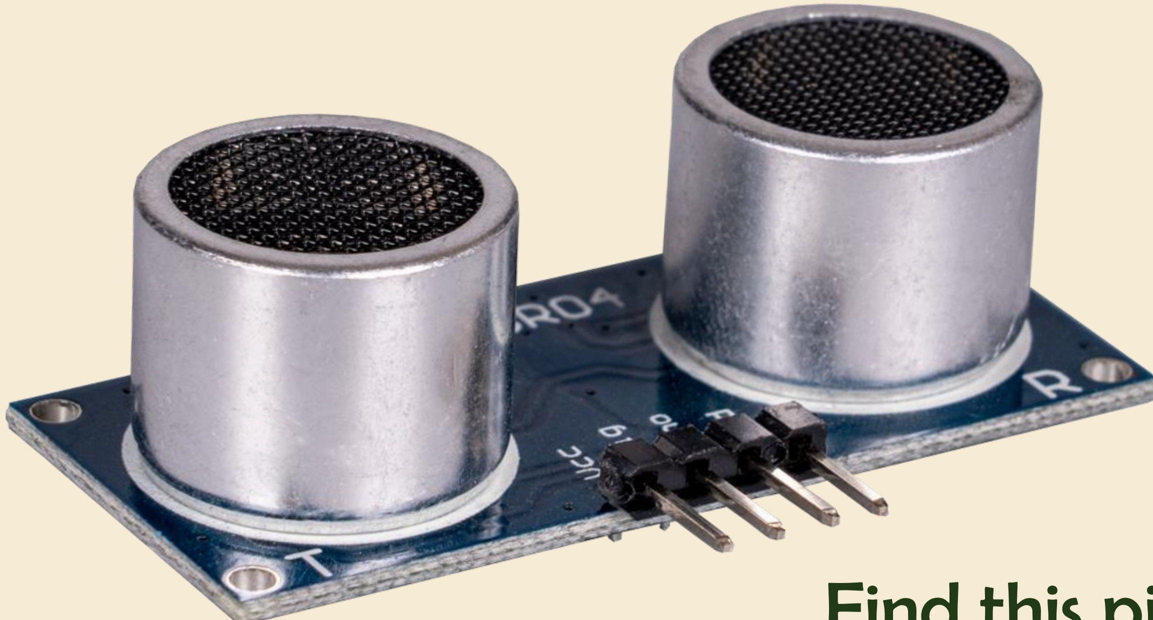
Sensors

Instruction Note:

- There will be a larger focus on sensors in lesson 2, the following slide will only give a brief overview.

Example Script:

- *"Sensors are the parts of machines (including robots) that tell them what is happening around them.*
- *There are many different types of sensors that can give a robot different information about their environment.*
- *Buttons or switches are the simplest types of sensor. A keyboard on a PC is also a sensor."*



Find this piece (1x)

What could it be for?

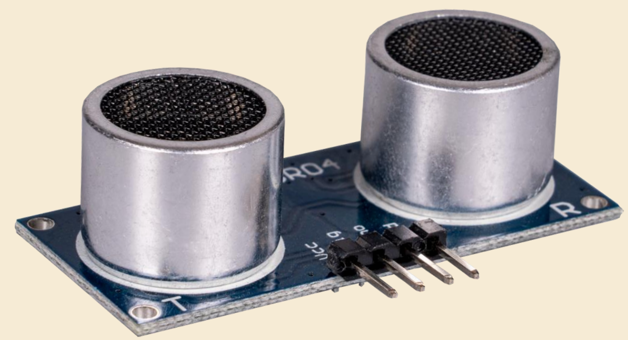
Sensors

- sensors are an 'input' device that can tell a robot what is happening around it

buttons

microphone

captures sound



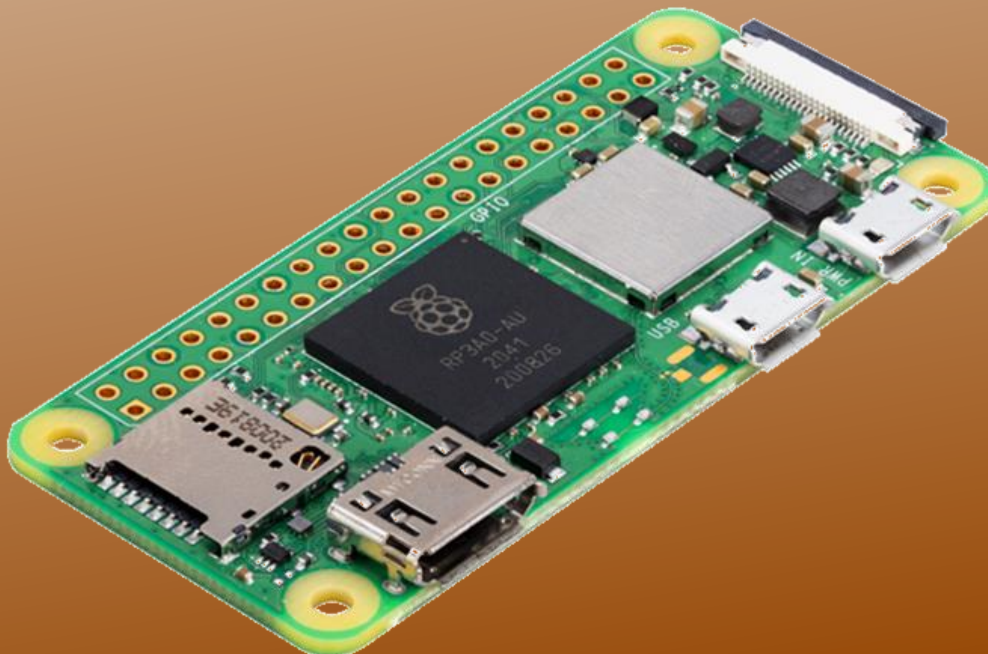
ultrasonic sensor

measures distance

Teacher-only slide – Content Extension

Raspberry Pi Zero 2W (Blossom's computer)

- While most computers come with a display to show user what is happening inside, some don't (like the Raspberry Pi, but it can be connected to a monitor and be used like any other computer, e.g. browsing internet)
- These so called Pi's are simple computers with less computing power but are still capable of many tasks a standard PC can perform.



Teacher-only slide – Instruction Guide

Computer

Instruction Note:

- Let students find the case of Blossom’s computer (the Raspberry Pi Zero) in the kit and think about what it could be for.
- Tell them they can look at it, but not take anything out.

Find this piece

What could it be for?

Teacher-only slide – Instruction Guide

Computer

Example Script:

- *"The computer is the control system of the robot.*
- *It controls everything the robot does, and can be seen as its brain.*
- *It runs code, human-written instructions telling the robot what to do.*
- *The wires coming out of the case later connect to the motors and the sensor, getting input data and sending signals to the motor, telling it when and how to move."*

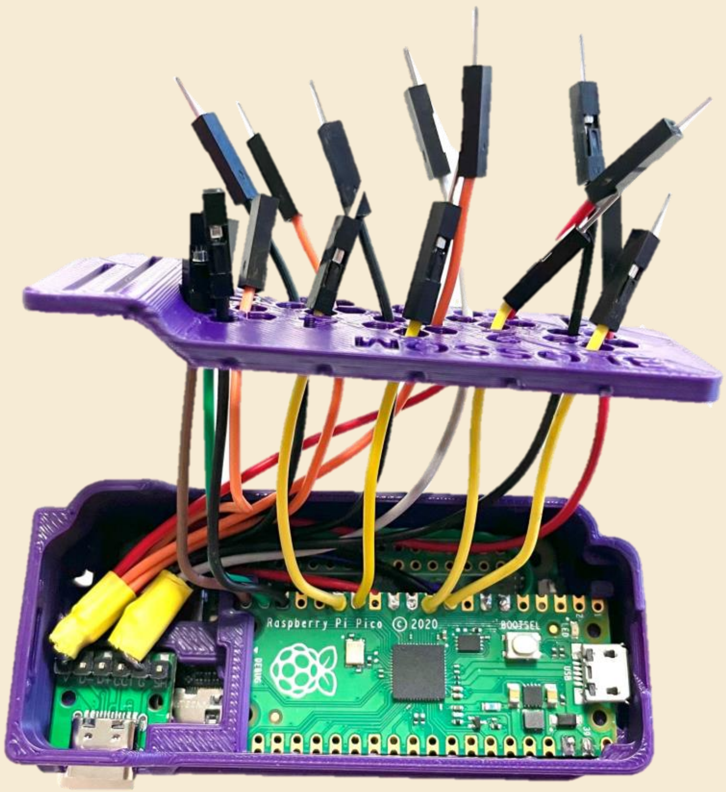
Instruction Note:

- There will be a larger focus on how computers process information and how we can tell them what to do with code in the next two lessons.

Computer

- The computer is the control system of a robot.
- It follows code (instructions that tell it what to do)

closed computer case



Teacher-only slide – Instruction Guide

Computers

Example Script:

- *“There are many kinds of computers. What computers do you know?*
  - possible answers: *laptops, tablets and smartphones*
  - *all robots you’ve seen earlier are controlled by a computer*
  - *many other machines also have a computer as a control structure inside them (e.g., digital cameras, dishwashers, washing machines, fridge, modern cars)*
- *On the next slide you can see*
  - *a smartphone, the inside of a smartphone,*
  - *a raspberry pi (the computer used inside of Blossom),*
  - *the case it is stored in when used for Blossom,*
  - *a laptop and the inside of a laptop.*

  These are all examples of what a computer can look like.”

Computer

- The computer is the control system of a robot.
- It follows code (instructions that tell it what to do)

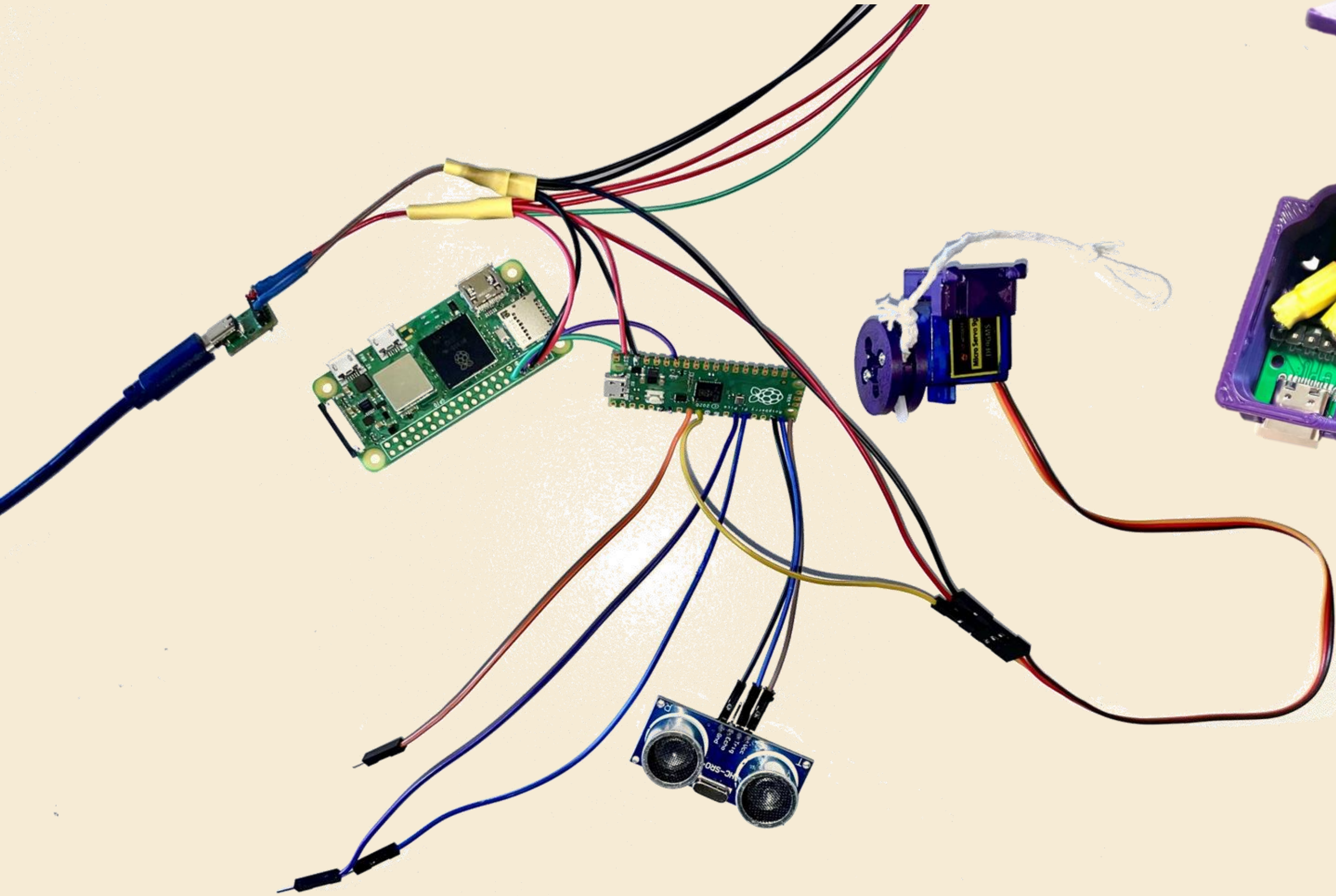
smartphone

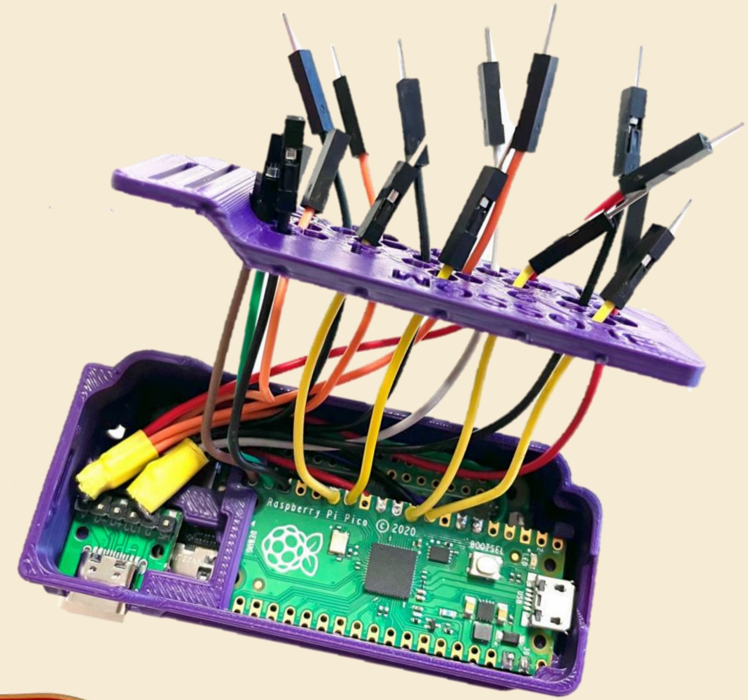
laptop

Blossom's computer

Wires

- connect the different parts
- transmit power and signals





Teacher-only slide – Instruction Guide

Control Process

Instruction Notes:

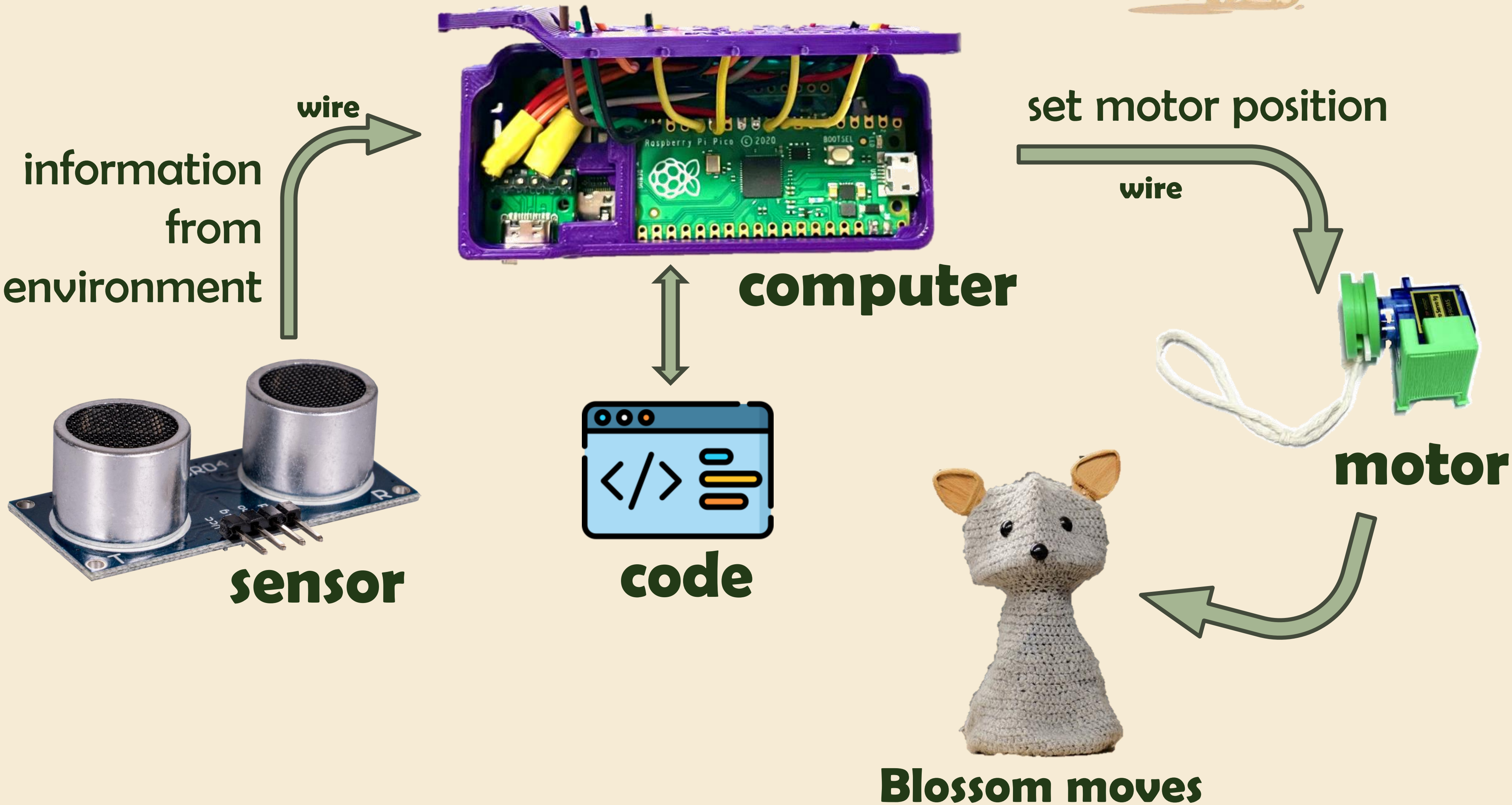
- The next slide shows in a simplified way how the different parts are connected and how environmental information is processed inside of Blossom (and many other robots and machines)
- If the previous parts of the lesson have taken longer than anticipated, you can skip this slide as it will be covered in more detail in lesson 2 & 3.

Example Script:

- *"The **sensor captures information from the environment** (in this case data on the distance of the next object above the sensor)"* → **more detailed content on this in lesson 2**
- *"Through **wires**, this information is **transmitted** to the raspberry pi zero 2w (the **computer** of Blossom). Using its software/**code**, it computes/**decides** the required **response**."* → **more detailed content in lessons 2 and 3**
- *"Since Blossom moves towards or away from an object in sensor mode, the response will be a movement, therefore the result of the decision-making is a **new motor position**. This information gets transmitted through other **wires** from the computer to the **motors**.*
- *In response, the motors move, which makes **Blossom** do the programmed **movement**."*

Control Process

OUTBACK ROBOTS
BUILD. LEARN. LEAP AHEAD.



Teacher-only slide – Instruction Guide

Connecting to Blossom and making a motor move

- The next slide contains a video & QR code for students to scan to get to the video with their own devices.
- The video explain how to access Blossom’s interface, how to connect the motor to the computer through wires, and how to make it move using the interface.

Connecting to Blossom and making a motor move

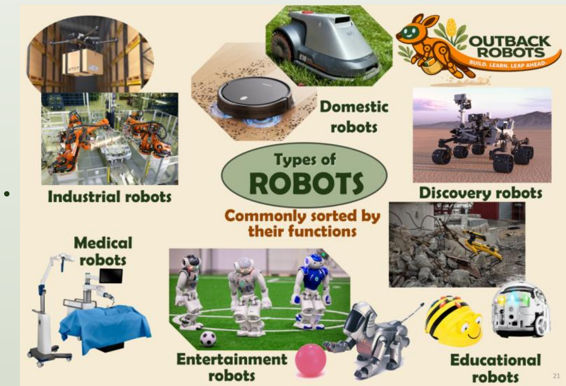
- Video
- QR code

Teacher-only slide – Instruction Guide

Discussion-
When should robots be used?

Instruction Note:

- Ask the students what they think when it is okay to use robots for a task?
  - You can do this activity while showing the adapted “Robot Types” slide. This way students can consider what different robot usages have in common. If you don’t want to use it, feel free to hide the slide.



- When might it be problematic?
  - Some **thoughts** on this can be found **on the next slide**. However, there is much more that can be said.
  - Further reads on ethical considerations around robot use: https://www.engineegroup.com/articles/TCSIT-7-156.php (general), https://link.springer.com/article/10.1007/s13347-019-00377-4 (workforce), https://www.oecd.org/content/.../Robots_and_AI-driven_robots_in_the_classroom_education.pdf (education)

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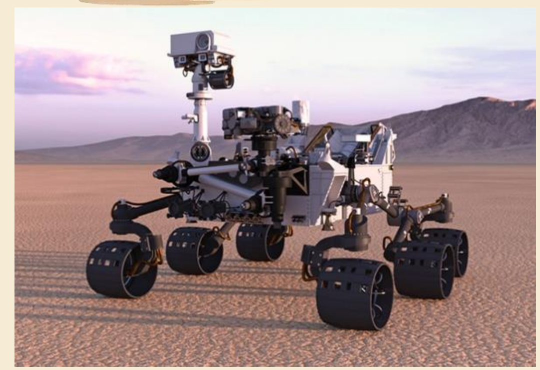
Discussion-
When should robots be used?

- **Can be useful:**
  - **dangerous**: for tasks or in places that could harm people (e.g. space exploration, search and rescue)
  - **dull**: long and repetitive tasks that humans might not want to do (e.g. domestic or industrial robots)
  - **precise**: for delicate tasks that are too difficult with our hands (e.g. medical, industrial)
  - **labour shortage**: tasks not enough humans are available for or willing to do
- **Can be problematic**:
  - **replacing humans**: job displacement & unemployment, isolation (e.g. if overreliance on robots in aged care), loss of human skilled labour
  - **befriending humans**: emotional dependency, illusion of caring but don't really understand what humans are saying (more in following lessons), can lead to social isolation, inappropriate responses to humans
  - **inequitable access**: robots often expensive, can require high technical knowledge to use
  - **deliberate misuse**: robots can be programmed to do harmful things

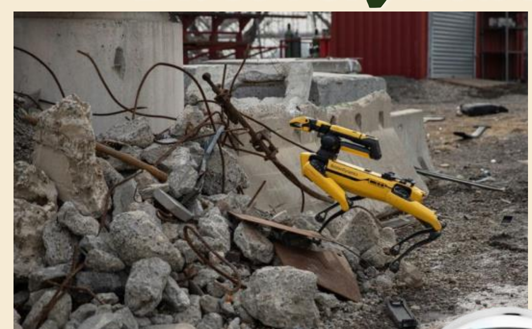
When should robots (not) be used?

Industrial robots

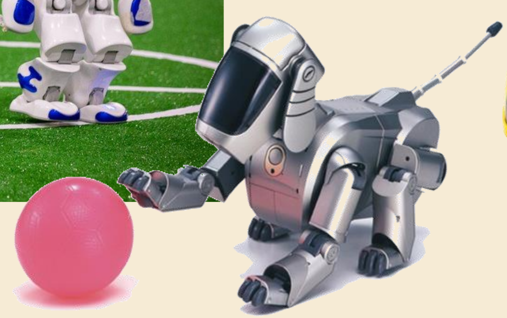
Domestic robots



Discovery robots



Medical robots



Entertainment robots

Educational robots

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Survey

- Lastly, let students answer the questionnaire for lesson 1.
- Tell them beforehand:
  - *"At the end of each lesson, you will all be asked to complete a short survey. The questions asked will be the same after each lesson."*
  - *[Note on how survey administered: paper vs online Qualtrics form link]*
  - *"This is **not a test**. There are no right or wrong answers when you're sharing your thoughts.*
    - *Your feedback will be used to help improve the lessons in the future.*
    - *It is important to answer honestly and to be as clear as possible."*
  - *"Try to respond to each question, but you can leave questions blank if you don't want to answer.*
  - *Don't share any personal information in your responses."*

Survey

- Instructions and picture/screenshot of the survey (whether paper or Qualtrics)

- Compare with experience in similar lessons.

Teacher-only slide – Instruction Guide

Summary

Go over the main points of the lesson again to help students consolidate their learning. You can let them talk with a partner first, then let one or two students answer to the whole class. There will always be a lesson recap at the beginning of the next lesson. If you are out of time after the survey, you can skip the summary.

- *"We learned today that there are many different types of robots and various tasks that robots are used for. Talk to the person next to you and see how many you can remember."*
  - Then ask the class to mention (at least) 3 different types/applications.
- *"We also saw that it's not always easy to tell robots from other types of machines, but there are some key features that most robots have in common. Discuss with the person next to you what makes a robot a robot.*
  - let students repeat the key features of robots (machines that use sensors to receive information from their environment, can move in the real world and are intelligent, i.e. can make decisions from the sensor input and the software/code to choose the next action)
- Some robots are social robots, repeat what that means (this is not a key learning objective and therefore not on the summary slide).
  - Social robots are those that have can have social and emotional communication abilities

Teacher-only slide – Instruction Guide

Summary

- Ask students to repeat the most important parts of a robot and what they are for.
  - ***sensors*** *(get information from environment),* ***motor*** *(creates movement),* ***computer + software*** *(controls what the robot does),* ***wires*** *(connect everything),* ***power source*** *(provides electrical energy that the pieces need to function),* ***mechanical structure*** *(holds everything together and gives the robot its shape)*
- Go over the simple control process from Blossom's sensor input to movement. (Skip if out of time, as this will be in the next lesson as well)
  - **sensor** captures information from the environment, transmits info through **wires** to **computer**, uses **code** to compute what to do, sends information through **wire** to **motor** to tell it what position to change to, motor **moves** → Blossom moves
- Let students repeat 2 helpful and 2 harmful uses of robots.
  - can be **helpful**: tasks are dangerous, dull, require high precision, labour shortage
  - can be **harmful**: replacing humans, befriending humans, inequitable access, deliberate misuse

Summary –
Introduction to Robots

- Robots come in many shapes. Name 3 types of robots.
- What makes a robot a robot?
  - machine that senses + moves in the real world + is intelligent (uses code to choose an action)
- Main robot parts
  - sensor (notice), motor (move), computer (control), wires (connect), power supply (energy), mechanical structure (holds together)
- Control Process from sensor input to movement:
  - sensor input → wire → computer with code → wire → motor
- Robots can be both helpful and harmful depending on the context. Name 2 examples for each.